Randomized Algorithm

Probability

1. (Discrete) random variable
2. Expectation

Q: "Coins ⇒ p heads, (1-p) tails. How many flips → See first Head?"

"Waiting Time Problem"

※ Showing that for $p \in (0,1)$

$$\sum_{i=1}^{\infty} i(1-p)^{i-1} = \frac{1}{p^2}$$

Set $x = 1-p$. $x \in (0,1)$ let $k = i-1$

$$S = \sum_{i=1}^{\infty} i(1-p)^{i-1} = \sum_{i=1}^{\infty} i\, x^{i-1} = \sum_{k=0}^{\infty} (k+1)x^k = 1 + 2x + 3x^2 + 4x^3 + \cdots$$

$$\Rightarrow S = 1 + 2x + 3x^2 + 4x^3 \cdots$$

$$xS = \quad x + 2x^2 + 3x^3 + \cdots$$

$$\Rightarrow (1-x)S = \underbrace{1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}}_{\text{Geometric Series!}}$$

$$\Rightarrow S = \frac{1}{(1-x)^2} = \frac{1}{p^2}$$

Let Waiting Time Be random variable T:

$$E(T) = \sum_{i=1}^{\infty} i \cdot P(T) = \sum_{i=1}^{\infty} i \cdot \underbrace{(1-p)^{i-1}}_{i-1 \text{ times tail}} \cdot \underbrace{p}_{\text{late time head}}$$

$$= p \cdot \frac{1}{p^2} = \frac{1}{p}$$

$$E(X+Y) = E(X) + E(Y)$$

$E(X \cdot Y) = E(X)E(Y)$ ONLY When X, Y are independent

Coupon Collector

n different type of coupon

how many boxes needs to be open to collect all kinds of coupon

Stage i → having i Coupon. want to have $i+1$ Coupon.

probability: $\frac{n-i}{n}$

let X_i be number of boxes to open to get the extra coupon

$$E(X_i) = \frac{1}{\frac{n-i}{n}} = \frac{n}{n-i}$$

(See waiting time problem)

let X be the number of boxes to open to get all coupons.

$$E(X) = \sum_{i=1}^{n} E(X_i) = \sum_{i=0}^{n-1} \frac{n}{n-i}$$

$$= 1 + \frac{n}{n-1} + \frac{n}{n-2} \cdots + \frac{n}{1}$$

$$= n \cdot \underbrace{\sum_{i=0}^{n-1} \frac{1}{i}}_{\text{"}\Theta(\log n)\text{"}} = \Theta(n \log n)$$

Guessing with no memory

$E(X_i) = \frac{1}{n}$ (n cards)

$$E(X) = \sum_{i=1}^{n} E(X_i) = 1$$

Guessing with memory

$$E(X_i) = \frac{1}{n-i+1}$$

$$E(X) = \sum_{i=1}^{n} E(X_i) = \frac{1}{n} + \frac{1}{n-1} + \cdots + 1 = \Theta(\log n)$$

Hiring Problem

```
Hire-Assistant(n):
best ← 0
for i ← 1 to n:
    interview Candidate i
    if Candidate i is better than best then:
        ☆ hire Candidate i
        best ← i
```

How many people are hired in Worse Case:

⇒ $\underline{\underline{n}}$ (all people)